

Name.....

Roll no.....

Physical Group

Far Western University
Faculty of science and Technology
B.Sc. Entrance Examination 2077

Model Questions

Full marks: 100

Time: 2 hrs

Answer all questions

Group A (Mathematics)

1. If $A = [-1, 3]$ and $B = [2, 4]$, then $A-B$ is equal to

(a) $[-1, 2]$ (b) $[-1, -2]$ (c) $[-1, 2]$ (d) $[-1, 3]$

2. The area of $\triangle ABC$ is,

(a) $\frac{1}{2} a \sin B \sin C$ (b) $\frac{1}{2} bc \sin A$ (c) $\frac{1}{2} a \sin A \sin C$ (d) $\frac{1}{2} abc$

3. If $1+6+11+\dots+x = 148$, then x is equal to

(a) 63 (b) 36 (c) 8 (d) None

4. The roots of the equation $\begin{vmatrix} x-1 & 1 & 1 \\ 1 & x-1 & 1 \\ 1 & 1 & x-1 \end{vmatrix} = 0$ are

(a) 1, 2 (b) -1, -2 (c) 1, -2 (d) -1, 2

5. The area of the triangle formed by the lines $y^2 - 9xy + 18x^2 = 0$ & $y = 9$ is

(a) $\frac{27}{2}$ sq. unit (b) $\frac{27}{4}$ sq. unit (c) 27 sq. unit (d) 54 sq. unit

6. The number of tangents to the circle $x^2 + y^2 - 2x + 4y = 0$ through $(-1, 2)$ is

(a) 1 (b) 0 (c) 2 (d) 4

7. $\lim_{x \rightarrow \infty} \frac{\sin \frac{1}{2}}{\frac{1}{2}}$ is equal to

(a) 1 (b) ∞ (c) 0 (d) does not exist

8. The derivative of $3x^2$ is

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- (a) 6 (b) $3x$ (c) $6x$ (d) $6x + 1$

9. If the function $y = f(x)$ is increasing function then the derivative of it is

- (a) zero (b) positive (c) negative (d) none

10. The different digits numbers between 2000 and 3000 can be formed from the integers 2, 3, 4, 5, and 6 is equal to

- (a) 20 (b) 21 (c) 23 (d) 24

11. The sum of infinite series

$$\frac{1.2}{1!} + \frac{2.3}{2!} + \frac{3.4}{3!} + \dots \text{ is}$$

- (a) $4e$ (b) $3e$ (c) $e + e^{-1}$ (d) $2e$

12. The eccentricity of hyperbola, $3x^2 - 4y^2 - 36$ is

- (a) $\frac{7}{2}$ (b) $\frac{\sqrt{7}}{4}$ (c) $\frac{\sqrt{7}}{2}$ (d) $\frac{\sqrt{7}}{5}$

13. The angle between the planes $2x - y + z = 6$ and $x + y + 2z = 3$

- (a) $\frac{\pi}{3}$ (b) $\frac{2\pi}{3}$ (c) $-\frac{\pi}{6}$ (d) $\frac{\pi}{6}$

14. The value of scalar product $\vec{a} \cdot \vec{b}$ is

- (a) $ab \sin \theta \cdot \cos \theta$ (b) $\frac{1}{2} ab \sin \theta$ (c) $ab \sin \theta$ (d) $ab \cdot \cos \theta$

15. The mean of the frequency distribution having C.V. = 5 and s. d. (σ) 2 is

- (a) 40 (b) 45 (c) 50 (d) 30

16. The probability of a card drawn at a random from well shuffled deck of 52 cards will be a heart is

- (a) $\frac{1}{3}$ (b) $\frac{13}{36}$ (c) $\frac{1}{2}$ (d) $\frac{1}{4}$

17. The set of $G = \{1, w, w^2\}$ is

- (a) additive group (b) Multiplicative group (c) both a and b (d) None

18. The coefficient of correlation 'r' lies between the range

- (a) $-3 \leq r \leq 3$ (b) $-1 \leq r \leq 1$ (c) $-1 < r < 1$ (d) $-1 > r > 1$

19. The distance covered by a body moving with a velocity of 360 km/hr has the uniform

acceleration 40 m/sec^2 in $\frac{1}{2}$ minute is

- (a) 21 m (b) 210 km (c) 21 km (d) 2100 m

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20. The K.E. of the body of mass 12 kg thrown vertically upwards with a velocity of 50 ms^{-1} is

- (a) 1500 J (b) 150000J (c) 15000J (d) 15 J

21. The perpendicular distance of the points (2,3,-5) from the plane $x+2y-2z=9$ is

- a) $\sqrt{3}$ b) $\sqrt{3}/4$ c) $\sqrt{3}/2$ d) 3

22. Which of the following are direction cosines?

- a) 1,0,0 b) 2,0,0 c) 1,1,1 d) $1/2, 1/2, 1/2$

23. The conic section is said to be hyperbola if

- a) $e=1$ b) $e>1$ c) $e<1$ d) $e\geq 1$

24. Find the sum of the series $16+8+4+2+\dots$ is

- a) 22 b) 32 c) 36 d) 40

25. If w is the cube root of unity then the value of $1+w+w^2$ is

- a) 1 b) 0 c) 2 d) 3

26. The perpendiculars drawn from the vertices on their opposite sides of triangle meet at a point which is called

- a) centroid b) incenter c) circumcenter d) orthocenter

27. The equation $\{x^2-2x+4=0\}$ where x is a real number is

- a) null set b) finite set c) singleton set d) infinite set

28. The middle term in the expansion of $(1+x)^{12}$ is

- a) 6^{th} b) 7^{th} c) 8^{th} d) none

29. The identity element of any group is

- a) Unique b) not exist c) 0 d) 1

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30. Which of the following function is discontinuous

- a) $2x+3$ b) x^2-2x+3 c) $1/x$ d) none

Group B (Physics)

31. Transformer works on the principle of

- (a) Lenz law (b) Tangent law (c) mutual induction (d) Biot-Savart law

32. To remove short sightedness, we use

- (a) converging lens (b) convex mirror (c) diverging lens (d) concave mirror

33. Photoelectric cell converts

- (a) electrical energy into heat energy (b) heat energy into electrical energy
(c) light energy into heat energy (d) none of these

34. Fleming's left hand rule is used to find out direction of

- (a) force on charged particle (b) force on neutral particle
(c) induced e.m.f. (d) velocity of charged particle

35. Brewster's law is related with

- (a) diffraction (b) interference (c) polarisation (d) diffraction and polarisation

36. Newton's law of cooling is valid for

- (a) solid (b) liquid (c) gas (d) liquid and gas

37. The example of non-ohmic resistance is

- (a) copper (b) transistor (c) silver (d) gold

38. When a ray of light enters a glass slab from air

- (a) its frequency remains constant (b) its wavelength increases
(c) its frequency increases (d) its frequency decreases

39. A solid ball of metal having concentric spherical cavity within it is heated, the volume of the cavity will

- (a) increase (b) decrease (c) remains constant (d) none of these

40. Amount of work done by centripetal force during circular motion of body is

- (a) positive (b) negative (c) zero (d) none of these

41. A man weighing mg standing inside lift moving upward with acceleration $4g$ has apparent weight

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- (a) Zero (b) 4mg (c) mg (d) 5mg

42.is not a unit of work.

- (a) Newton meter (b) Joule (c) erg (d) Joule per second*

43. For zero range, projectile should fired at an angle of

- (a) 30^0 (b) 45^0 (c) 60^0 (d) 90^0

44. Distance covered by body in n^{th} second is

- (a) $u + \frac{a}{2}(2n-1)^{**}$ (b) $u + \frac{a}{2}(2n+1)$ (c) $u - \frac{a}{2}(2n-1)$ (d) $u - \frac{a}{2}(2n+1)$

45. If momentum of body is increased by 50% then percentage change in its kinetic energy is

- (a) 25% (b) 75% (c) 100% (d) 125%

46. Resistance of metallic wire is directly proportional to its

- (a) cross sectional area (b) length (c) volume (d) surface area

47. Surface tension is the property of

- (a) solid and liquid (b) liquid and gas (c) gas and solid (d) only liquid

48. Persistence of hearing is

- (a) 1s (b) 0.1s (c) 10s (d) 0.01s

49. Isochoric process takes place at constant

- (a) temperature (b) pressure (c) volume (d) pressure and volume

50. The resonance frequency of L-C-R series circuit is

- (a) $\frac{1}{2\pi\sqrt{LC}}$ (b) $\frac{\pi}{\sqrt{LC}}$ (c) $\frac{\sqrt{2\pi}}{LC}$ (d) $\frac{1}{2\pi}\sqrt{\frac{C}{L}}$

51. A stone is dropped from a certain height which can reach ground in 5 seconds it is stopped after 3 sec of its fall and then again released ,the total time taken by the stone to reach the ground will be

- a) 6 sec b) 6.5 sec c) 7 sec d) 7.5 sec

52. An airplane moving with 300km/hr in circular path then what is change in velocity in half revolution ?

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- a) zero b) 300Km/hr c) 900Km/hr d) 600km/hr

53. For a given material ,the Young's modulus is 2 times that of rigidity modulus,its poisson ratio is

- a) 1 b) 0 c) 0.2 d) 0.5

54. If the earth suddenly contract $\frac{1}{6}$ of present size without change in mass the duration of day will be nearly

- a) 0.67 hours b) 5 hours c) 6 hours d) 4 hours

45. How much a water a pump of 20 KW can raise in one minute to a height of 10 m

- a) 10000 litres b) 12000 litres c) 1000 litres d) 2000 litres

56. When two electrons brought near the potential energy of the system

- a) Increases b) Decreases c) Remains constant d) zero

57. The ratio of specific charge of proton to that alpha particle is

- a) 1:2 b) 2:1 c) 1:1 d) 4:1

58. Magnetic permeability of diamagnetic substance is

- a) Positive b) Negative c) positive and high d) zero

59. X-rays are

- a) Electromagnetic wave b) Highly energetic c) Absorbed by metal d) All of above

60. Zener diode is used for

- a) Rectifier b) Amplification c) Stabilization d) All of above

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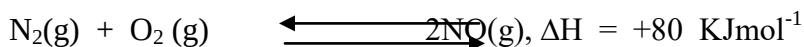
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Group C (Chemistry)

61. Which of the following weighs least?
- (a) 2.0 gram mole of CO_2 (b) 0.1 gram of sucrose($\text{C}_{12}\text{H}_{22}\text{O}_{11}$)
- (c) 1.0 gram atom of calcium (d) 1.5 mole of water
62. The temperature of a gas in closed container is 27°C . if the temperature is raised to 327°C , the pressure exerted is :
- (a) reduced to half (b) doubled
- (c.) reduced to one third (d.) can't be calculated from the above information
63. The spectral line in hydrogen spectrum obtained when the electron jumps from $n = 5$ to $n = 2$ energy level belongs to
- (a) Lyman series (b) Balmer series (c) Paschen series (d) Pfund series
64. The property which regularly increases down the group in the periodic table is
- (a) Ionization energy (b) Electronegativity (c) Reducing nature (d) Electron nature
65. Which of the following molecule is not an exception to octet rule
- (a) CO_2 (b) BF_3 (c) PF_5 (d) IF_7
66. The formation of an adduct between NH_3 and BF_3 involves the formation of
- (a) Ionic Bond (b) co-valent bond
- (c) Co-ordinate co-valent bond (d) Hydrogen bond
67. Which of the following have the highest boiling points ?
- (a) HCl (b) HF (c) HBr (d) HI
68. ΔH for the combustion of a compound is
- (a) Positive (b) Zero (c) Negative (d) May be positive or negative
69. The heat of neutralization of a strong acid by a strong base is
- (a) 13.7Kcal (b) $>13.7\text{Kcal}$ (c) $<13.7\text{Kcal}$ (d) none of these
70. The p^{H} of 10^{-8}M solution of HCl is
- (a) 8 (b) 6 (c) between 6 and 7 (d) between 7 and 8
71. Nitrogen combines with oxygen to form nitric oxide;

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The decomposition of NO gas is favoured by

- (a) decrease in the pressure (b) increase in pressure
(c) Decrease In Temperature (d) increasing the concentration of N_2

72. During the electrolysis of fused NaCl, which reaction takes place at anode?

- (a) Na^+ ions are oxidized (b) Cl^- ions are oxidized
(c) Na^+ ions are reduced (d) Cl^- ions are reduced

73. The role of a catalyst in a chemical reaction is to change;

- (a) heat of reaction (b) final products
(c) activation energy (d) equilibrium constant

74. Oxygen and ozone are;

- (a) allotropes (b) isomers (c) isotopes (d) isobars

75. which of the following metals give hydrogen gas with very dilute HNO_3 ;

- (a) Zn (b) Fe (c) Sn (d) Mn

76. The IUPAC name of the following compound

$\text{CH}_3\text{COCH}_2\text{CH}_2\text{OH}$ is;

- (a) 3-oxobutan-1-ol (b) 1-hydroxybutan-3-one
(c) 4-hydroxybutan-2-one (d) 2-oxobutan-4-ol

77. The ozonolysis of $(\text{CH}_3)_2\text{C} = \text{C}(\text{CH}_3)_2$ gives following compound ;

- (a) acetone (b) acetaldehyde and acetone (c) acetic acid (d) formic acid

78. Benzene react with acetyl chloride in the presence of anhydrous AlCl_3 to give

- (a) toluene (b) acetophenone (c) benzophenone (d) benzoic acid

79. which of the following base is formed only in RNA and not in DNA?

- (a) uracil (b) thymine (c) guanine (d) adenine

80. Freon-12 is commonly used as

- (a) an insecticide (b) a refrigerant (c) a solvent (d) fire extinguisher

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93. You should work much than that

- a) hardly b) more hardly c) harder d) more harder

94. I'd rather you.....the dinner now.

- (a) cook (b) will cook (c) didn't cook (d) have cooked.

95. He dances wonderfully

- a) inspite of he has a bad leg. b) inspite his bad leg.
c) inspite he has a bad leg d) inspite of his bad leg

96. He didn't sing, and I didn't sing.....

- (a) also (b) too (c) neither (d) either

97. " Would you mind if I sat here ?" is

- a) question b) Statement c) apology d) permission

98. A stitch in time saves.....

- (a) time (b) nine (c) life (d) fine

99. The correct sentence from the prompts the building/build/2008 is

- a) The building has been built in 2008. b) The building was built in 2008
c) The building had been built in 2008 d) The building was built until 2008

100. The word 'phone' has the same vowel sound as ;

- (a) lawn (b) globe (c) done (d) fan